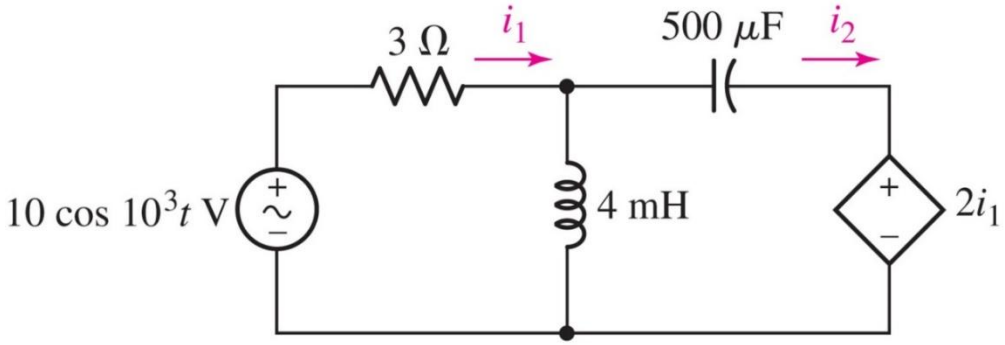
Exercise 9.1: Apply Mesh Analysis using Phasors

- Find the currents $i_1(t)$ and $i_2(t)$ through mesh analysis of phasors. ***Express the answers in both phasor/frequency and time domain.***



Answer:

$\mathbf{I}_1 = 1.24\angle 29.7^o$ A, $i_1(t) = 1.24\cos(10^3t + 29.7^o)$ A

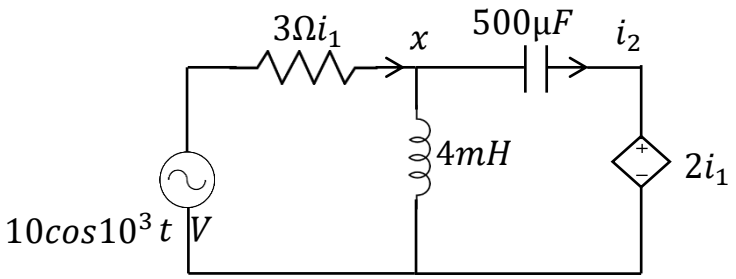
$\mathbf{I}_2 = 2.77\angle 56.3^o$ A, $i_2(t) = 2.77\cos(10^3t + 56.3^o)$ A

Solution for Exercise 9.1 (1/2)

Analysis:

- Mesh requires loops.
- Recall phasors do not have time "*t*" expression. Questions ask for both **I** and "*i*(*t*)".

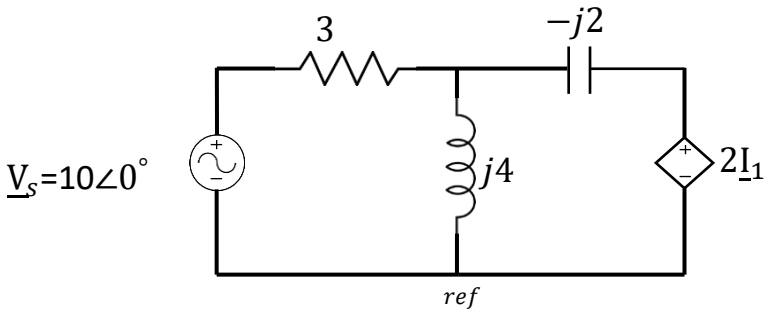
Circuit in time domain:



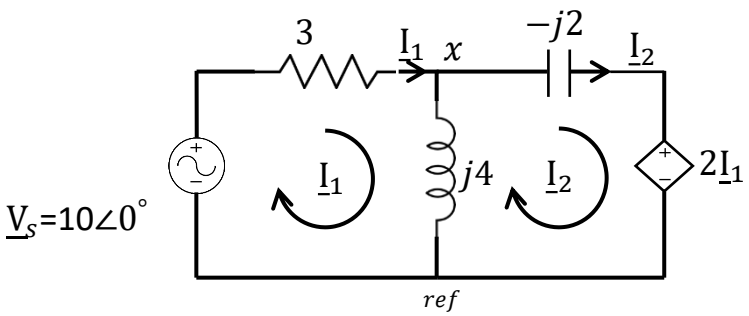
$$j\omega L = j(10^3)(4m) = j4$$

$$\frac{1}{j\omega c} = \frac{1}{j(10^3)(500\mu)} = -j2$$

Convert to phasor domain:



Solution for Exercise 9.1 (2/2) – Mesh Analysis



Mesh 1: $\underline{V}_s = 3\,\underline{I}_1 + \mathrm{j}4(\underline{I}_1 - \underline{I}_2)$

Mesh 2: $-2\underline{I}_1 = j4(\underline{I}_2 - \underline{I}_1) - j2\underline{I}_2$

$$\underline{I}_1 = \underline{I}_2 . \left(\frac{j2}{j4-2}\right)$$

$$\underline{V}_s = (3+j4).\underline{I}_2 . \left(\frac{j2}{j4-2}\right) - j4\underline{I}_2$$

$$= \left(\frac{-8+j6}{j4-2} - j4\right)\underline{I}_2 = \left(\frac{8+j14}{-2+j4}\right)\underline{I}_2$$

$$\underline{I}_2 = \frac{-2+j4}{8+j14} . \underline{V}_s = \frac{-2+j4}{8+j14} \quad (10\angle 0°)$$

$$\underline{I}_2 = 2.77\angle 56.3°$$

$$\underline{I}_1 = (2.77\angle 56.3°)\left(\frac{j2}{j4-2}\right)$$

$$\underline{I}_1 = 1.24\angle 29.7° \text{ A}$$

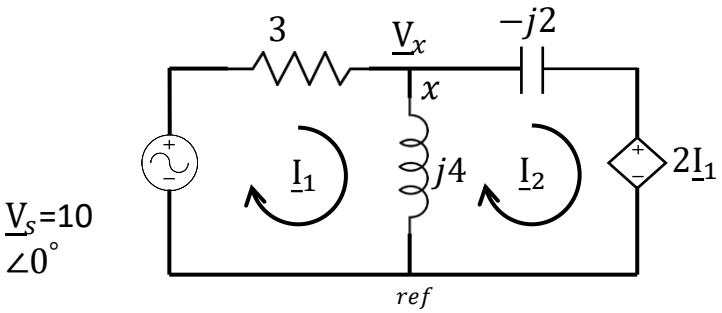
Intermediate answers, in phasors:
$\underline{I}_1 = 1.24\angle 29.7°$ A, $\underline{I}_2 = 2.77\angle 56.3°$.

Answer required are in time domain:

$$i_1(t) = 1.24\cos(10^3 t + 29.7^o)\text{ A}$$
$$i_2(t) = 2.77\cos(10^3 t + 56.3^o)\text{ A}$$

If asked by the question, mesh analysis can also be used to determine the voltage function/phasor at point x. For example, $\underline{V}_x = j4(\underline{I}_1 - \underline{I}_2) = ??\text{V}$

Solution for Exercise 9.1 (extra) – Nodal Analysis



Use nodal analysis: $\frac{\underline{V}_s-\underline{V}_x}{3}=\frac{\underline{V}_x}{j4}+\frac{\underline{V}_x-2\underline{I}_1}{-j2}$

$$j4(\underline{V}_s-\underline{V}_x)=3\underline{V}_x+\frac{\underline{V}_x-2\underline{I}_1}{-j2}(j4)(3)$$

$$j4(\underline{V}_s-\underline{V}_x)=3\underline{V}_x+12I_1-6\underline{V}_x$$

$$j4(\underline{V}_s-\underline{V}_x)=-3\underline{V}_x+12(\frac{\underline{V}_s-\underline{V}_x}{3})$$

$$(7-j4)\underline{V}_x=(4-j4)\underline{V}_s$$

$$\underline{V}_x=\underline{V}_s.\frac{4-j4}{7-j4}=(10\angle 0°)\left(\frac{4-j4}{7-j4}\right)$$

$$\underline{V}_x=7.02\angle -15.3°\text{V}$$

Exercise 9.1 asks for i_1, i_2:

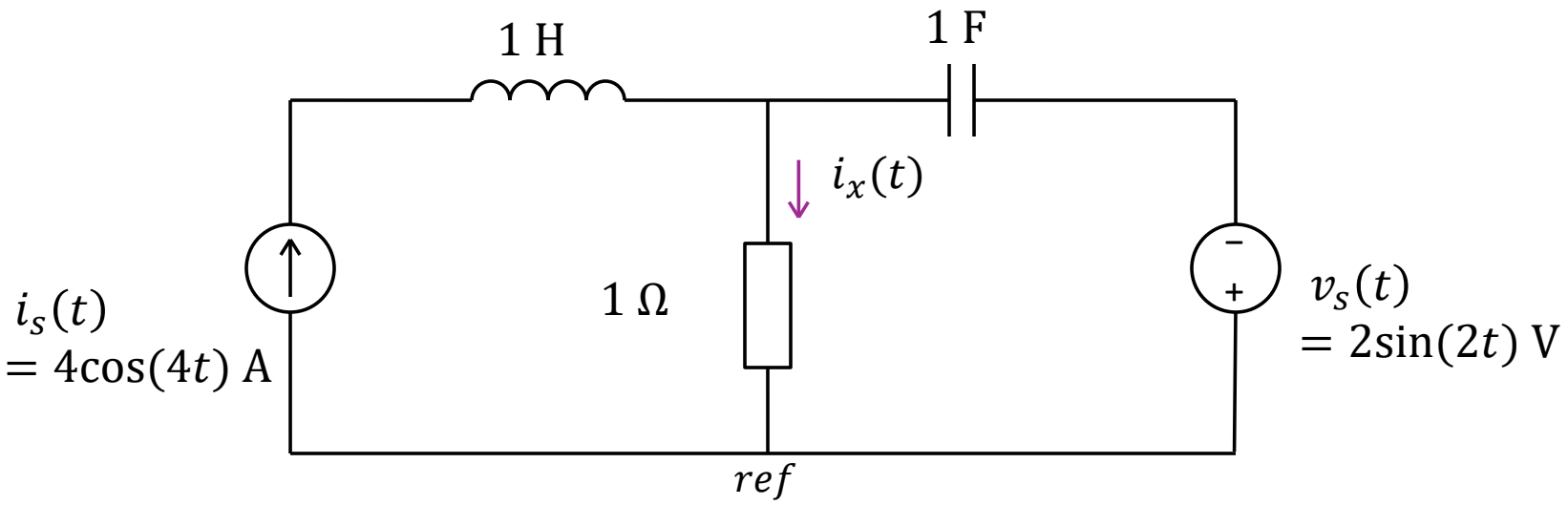
$$\underline{I}_1=\frac{10\angle 0°-7.02\angle -15.3°}{3}=1.24\angle 29.7°\ \text{A}$$

$$\underline{I}_2=\frac{7.02\angle -15.3°-2(1.24\angle 29.7°)}{-j2}$$

$$=2.77\angle 56.3°\ \text{A}$$

Exercise 9.2: Superposition at different frequencies

- Determine the downward current through the 1 Ω resistor using phasor analysis and superposition. Assume sinusoidal steady state.



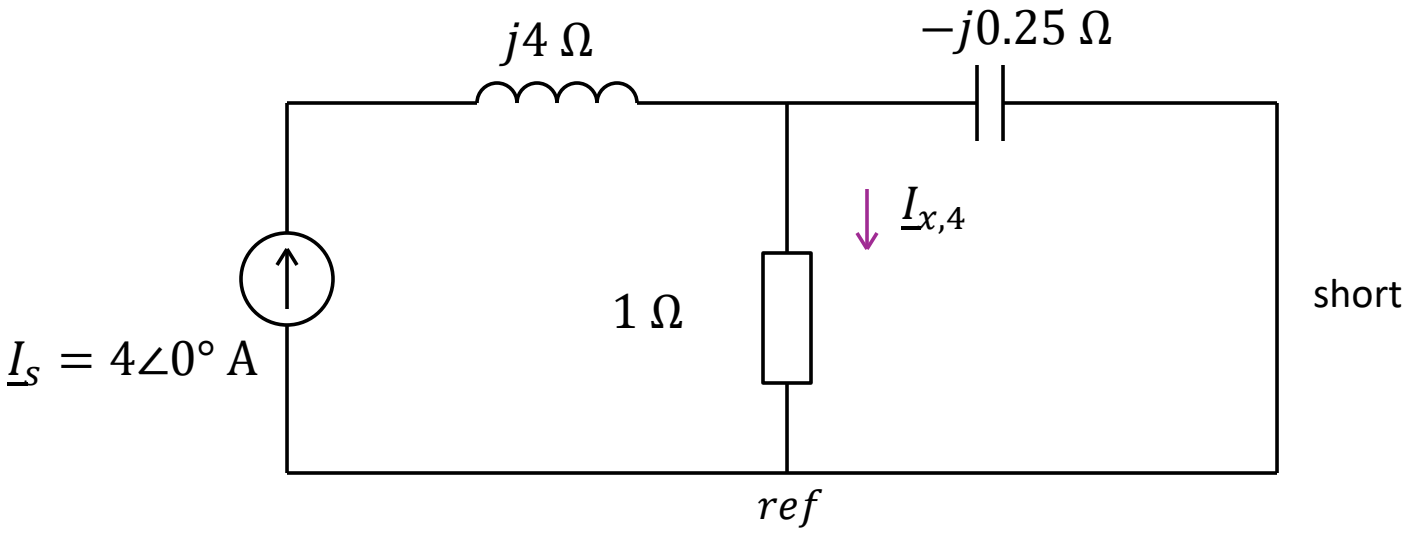
$$i_x(t) = 0.9701\cos(4t - 75.9638°) + 1.7889\cos(2t + 116.5651°)\ \text{A}$$

Solution for Exercise 9.2: Current-source contribution

Analysis:

- Use peak phasors with a cosine reference. Show each source's contribution, then give the total time-domain current.
- Explain why the two current phasors cannot be added directly.

➢ Activate the current source at 4 rad/s. Replace the voltage source by a short circuit.



$$Z_C = \frac{1}{j(4)(1)} = -j0.25\ \Omega$$

$$\underline{I}_{x,4} = 4\frac{1}{1+j4} = 0.2353 - j0.9412$$

$$\underline{I}_{x,4} = 0.97\angle -76.0°\ \mathrm{A}$$

Solution for Exercise 9.2: Voltage-source contribution

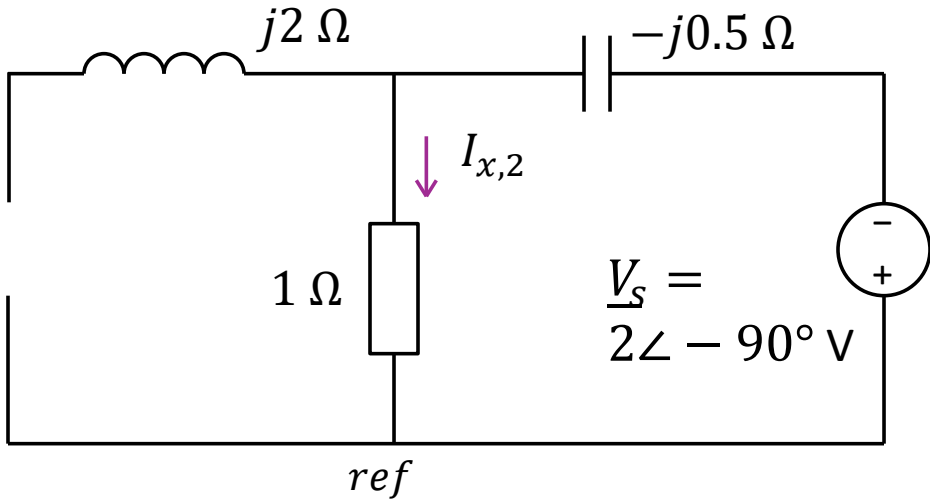
Activate the voltage source at 2 rad/s. Open the current-source branch, so the inductor branch carries no current. Arbitrarily, we also use "cosine" as the reference, just to be consistent with the choice for the current source (at different frequency):

$$v_s(t) = 2\sin(2t) = 2\cos(2t - 90°)\ \text{V}$$

The source's upper terminal is negative. Its upper-node voltage relative to the bottom conductor is therefore:

$$\underline{V_s} = 2\angle -90° = -j2\ \text{V}$$

$$Z_C = \frac{1}{j(2)(1)} = -j0.5\ \Omega$$



$$\underline{I_{x,2}} = \frac{-\underline{V_s}}{1 + Z_C} = \frac{j2}{1 - j0.5} = \frac{2\angle 90^o}{1.118\angle -26.6°}$$
$$= 1.79\angle 116.6^o = -0.8 + j1.6\ \text{A}$$

Solution for Exercise 9.2

$$i_{x,4}(t) = 0.97\cos(4t - 76.0°)\ \text{A}$$
$$i_{x,2}(t) = 1.79\cos(2t + 116.6°)\ \text{A}$$

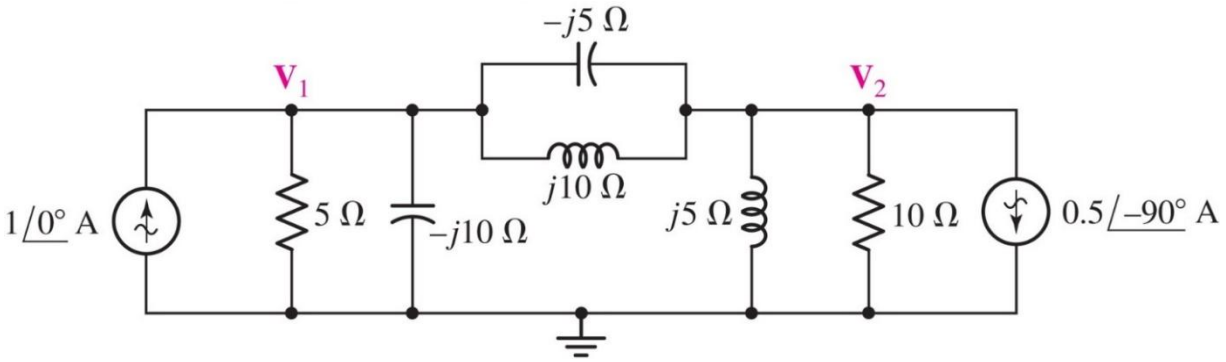
Add the two responses in the time domain:

$$i_x(t) = 0.97\cos(4t - 76.0°) + 1.79\cos(2t + 116.6°)\ \text{A}$$

- The frequencies differ, so there is no single phasor representing the total current.
- The ideal current source fixes its series-inductor current. This is why the inductance does not appear in the final resistor-current expression.

Exercise 9.3: Apply Thevenin's Theorem using Phasors

- Find $\mathbf{V}_1$ in the circuit below using Thevenin's equivalent circuit theorem.



Answer:

$\mathbf{V}_1 = 1 - j2$ V

$\mathbf{V}_2 = -2 + j4$ V

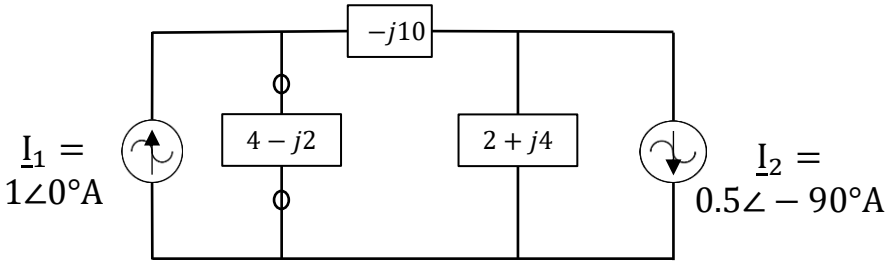
(if ever asked by the question)

Additional: Textbook example 10.8 - uses Nodal analysis.

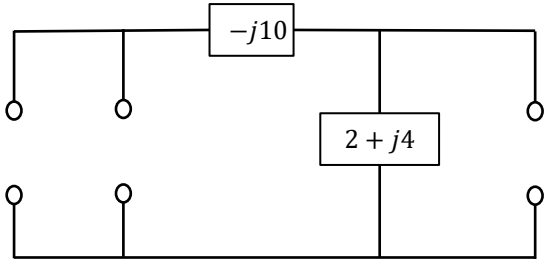
Solution for Exercise 9.3 (Thévenin method)

Analysis:

- There are several options on which part of the circuit to be simplified using TEC. Usually no single best answer. Choose the one that "makes sense". Always remember the objective of the problem: in this case, to find $\mathbf{V}_1$ phasor.

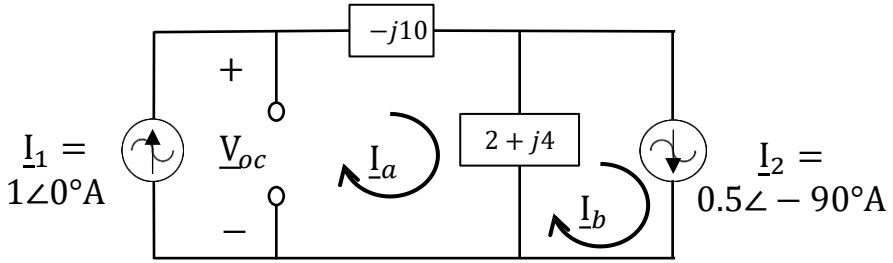


Calculate the Thévenin impedance:



$$\underline{Z}_{Th} = -j10 + (2+j4) = 2 - j6$$

Calculate the Thévenin voltage:



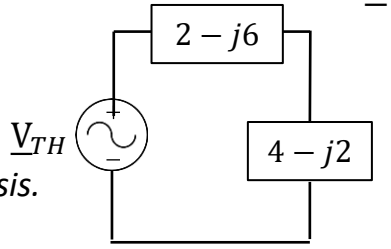
$$\underline{I}_a = \underline{I}_1 = 1\angle 0°\ \text{A}$$

$$\underline{I}_b = \underline{I}_2 = 0.5\angle -90°\ \text{A}$$

$$\underline{V}_{oc} = -j10\underline{I}_a + (2+j4)\ (\underline{I}_a - \underline{I}_b)$$

$$= -j10\ (1\angle 0°) + (2+j4)\ (1\angle 0° - 0.5\angle -90°)$$

$$= 5\angle -90°$$



$$\underline{V}_1 = \frac{4-j2}{2-j6+(4-j2)}\underline{V}_{TH} = (1-j2)\ \text{V}$$

Additional: Textbook example 10.8 - uses Nodal analysis.
$\mathbf{V}_2 = -2 + j4$ V *(if ever asked by the question)*